

IJTAG Gate-Level 通用除錯指南

TDO X、valid X、shadow event register、SDF timing 與 Tessent pattern completeness



本指南適用於已完成technology mapping的任意DUT. 假設gate netlist已對應到製程standard cells, 無GTECH/unmapped cells, 並使用RTL extraction時產生的ICL/PDL retarget patterns進行驗證.

第一原則

- TDO波形曾出現X不一定是錯誤; 必須判斷是否位於有效compare window.
- 先找第一個從0/1變成X的內部節點, 不要直接從top-level TDO猜測.
- 先比較RTL、gate/no-SDF、gate/SDF三組結果, 再決定查邏輯還是查時序.
- Tessent ICLNetworkVerify主要證明IJTAG access transport, 不會自動理解時序電路的valid語意.

最快的結果分類

結果	優先懷疑
RTL PASS; gate/no-SDF FAIL	reset、cell model、CDC、clear race、netlist mapping
gate/no-SDF PASS; gate/SDF FAIL	setup/hold、recovery/removal、pulse width、SDF corner/timescale
valid X但Tessent仍PASS	valid bit可能是don't-care或compare mask未開啟
shadow_valid正常; TDR bit X	Capture-DR edge、TDR input timing或bit mapping
TDR bit正常; TDO X	SIB/TAP mux、TDO enable、多重driver或sampling edge

1. TDO X 是否真的構成錯誤

先把TDO X分成三類.只有第三類一定要視為pattern failure.

X出現位置	判定	下一步
reset前、非Shift-DR、state transition附近	可能正常	確認compare enable為0
Shift-DR期間, 但expected/mask標示don't-care	simulation可能PASS	確認mask是合理需求, 不是把錯誤藏起來
有效sampling point且compare enable=1	真正錯誤	從TDO往內追第一個X

必看波形

```
tck, tms, trst  
tap_state / capture_dr / shift_dr / update_dr  
sib_select / active_scan_segment  
tdr_shift_out / internal_tdo / top_tdo  
expected_tdo / compare_enable / compare_mask
```

如果Tessent報告PASS但valid對應的TDO bit是X, 最先確認expected TDO與mask.自動network pattern可能只在驗證scan transport, 並未要求valid等於1.

常見TDO X來源

來源	典型現象
TAP未reset	instruction/state/select registers從X開始
TDR未完全初始化	前幾個shift bits或特定位元週期性出現X
SIB select為X	instrument path與bypass path經mux後變X
instrument DataIn為X	Capture-DR後X進入TDR, 再shift到TDO
standard-cell model不完整	unresolved module、UDP或power pin問題
SDF notifier	某個clock edge後DFF Q突然變X, log同時有timing violation
sampling race	只有TCK edge附近短暫X, 穩定取樣點可能正常

2. 三段模擬定位法



RTL	Gate no SDF	Gate + SDF	結論
PASS	FAIL/X	-	合成後邏輯、reset、CDC、library model或pattern/netlist對接
PASS	PASS	FAIL/X	SDF timing、notifier、clock/reset timing或corner
PASS	PASS	PASS但波形有X	檢查X是否只出現在masked/non-compare cycles
PASS	FAIL/X	FAIL/X	先不要分析SDF; gate functional問題已存在

建議執行順序

Run A: RTL + retargeted patterns
 Run B: gate netlist + full standard-cell Verilog models, no SDF
 Run C: same gate setup + SDF

Compare the first timestamp where valid or TDO becomes X.

Gate/no-SDF失敗時

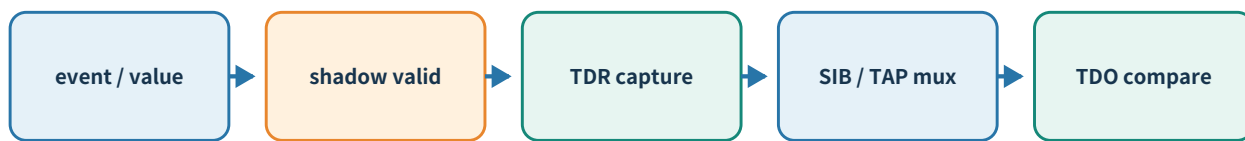
- 搜尋unresolved module、black box、UDP not found、power pin mismatch.
- 確認netlist中shadow register確實存在, 且沒有被constant removal或register merge.
- 確認reset與functional clock在gate testbench中真的有drive.
- 確認RTL pattern與gate top module、TAP pins及scan length一致.

只有SDF失敗時

- 找到第一個setup/hold/recovery/removal/pulse-width violation.
- 核對SDF instance hierarchy、timescale與min/typ/max corner.
- 確認reset解除遠離functional clock與TCK edge.
- 診斷時可暫時關閉timing checks/notifier, 但不能將該PASS當成最終結果.

valid是單cycle pulse或event時, 比長時間保持的value更容易因SDF delay、CDC或pulse width變成X.因此value正常不代表valid路徑也安全.

3. 從 valid 追到 TDO 的第一個 X



Trace the first 0/1 to X transition

建議在同一張waveform依序加入以下訊號, 找第一個不再是確定0/1的節點.

```
rst_n  
functional_clk  
tck  
event / valid_source  
shadow_valid_d / shadow_valid_q  
clear_shadow / read_ack  
capture_dr  
tdr_valid_capture_bit  
tdr_shift_out  
internal_tdo  
top_tdo  
compare_enable / compare_mask
```

第一個X的位置	問題範圍
event/valid_source	功能電路或event生成條件
shadow_valid_q	reset、setup/hold、CDC、set/clear overlap或cell model
TDR valid capture bit	Capture-DR timing、DataIn mapping或capture mux
TDR shift output	TDR initialization、shift length或bit order
internal_tdo/top_tdo	SIB/TAP mux、output enable或multiple drivers
只有compare結果錯	sampling edge、expected bit或mask設定

TDR資料模式除錯

- all-0/all-1: 找stuck-at或未初始化bit.
- 0xAA/0x55交錯: 找相鄰bit與bit-order問題.
- walking-1/walking-0: 確認每個TDR位置與總shift length.

4. Shadow event register 專項除錯

RTL加入shadow register只能保證event在邏輯上被保留; gate + SDF還必須滿足clock、reset、CDC及set/clear protocol.

常見失效機制

機制	Gate-level表現	修正方向
event太靠近clock edge	shadow Q由notifier變X	增加setup margin或同步event
reset解除靠近clock	recovery/removal violation	延後操作並避開clock edge
event與TCK不同domain	RTL正常, SDF後capture不穩	sticky flag + 2FF同步到TCK
event與clear同時發生	RTL依if優先序正常, gate產生race/X	明確握手; read完成後再clear
event本身當clock	pulse-width/glitch violation	使用真正clock + clock enable
shadow register被最佳化	netlist找不到或與其他reg合併	查DC log/netlist, 再做精準preservation
pattern capture太早	value有值但valid尚未同步	等待足夠functional/TCK cycles或寫application PDL

較穩健的跨clock架構

```
Functional clock domain          TCK domain

event -> sticky shadow_valid -> sync_ff1 -> sync_ff2 -> TDR capture
      ^                                     |
      |----- synchronized read acknowledge -----|
```

若clear_shadow來自TCK domain, 不要直接送進functional-clock always block.需要同步或toggle/handshake機制.最簡單的bring-up方式是先讓sticky valid只由reset清除, 排除clear CDC後再加入ack.

同clock domain的安全優先序

```
always @(posedge func_clk or negedge rst_n) begin
    if (!rst_n)
        shadow_valid <= 1'b0;
    else if (event)
        shadow_valid <= 1'b1;
    else if (clear_shadow_sync)
        shadow_valid <= 1'b0;
end
```

5. SDF與standard-cell model檢查

沒有GTECH只證明DC完成technology mapping; gate simulation仍依賴正確的standard-cell Verilog model、UDP、power pins與SDF hierarchy.

Simulation log搜尋

```
grep -Ei "unresolved|module not found|black.?box|UDP" sim.log
grep -Ei "setup|hold|recovery|removal|width|notifier" sim.log
grep -Ei "SDF.*error|instance.*not found|cannot annotate" sim.log
grep -Ei "shadow|valid" sim.log
```

檢查	成功條件
Cell model	所有DFF/mux/tie/clock-gating/UDP都有functional model
SDF hierarchy	SDF instance path與simulation top一致
SDF coverage	TAP/SIB/TDR與shadow register的timing arcs成功annotation
Timescale	testbench、cell model與SDF單位一致
Corner	明確記錄min/typ/max; max偏setup, min偏hold
Reset timing	assert足夠久; deassert遠離functional clock與TCK edge
TCK period	符合SDF corner下IJTAG path的實際能力

Notifier判定

- 若DFF Q在clock edge後立刻變X, 同時log有timing violation, 高機率是notifier.
- 若移除SDF後X消失, 不要修改ICL/PDL; 應修正timing、clock/reset sequencing或降低TCK頻率.
- 若關閉notifier後PASS, 只能用來證實X來源, 不能當作signoff結果.

SDF annotation有warning不一定失敗, 但若warning落在shadow valid、TDR capture、TAP state register或TDO path上, 必須逐項處理.

6. Tessent自動pattern完整性

Tessent Shell User's Manual說明, ICL-based verification會確認每個IJTAG

TDR可透過iWrite與iRead存取.這是access-network驗證, 不是instrument功能coverage.

```
set design_top design_top
set_context patterns -ijtag -design_id gate
read_design $design_top -design_id gate -verbose
set_current_design $design_top
check_design_rules

set spec [create_patterns_specification]
report_config_data $spec
process_patterns_specification

run_testbench_simulations
check_testbench_simulations -report_status
```

report_config_data應確認

```
Patterns(ICLNetwork) {
    ICLNetworkVerify(design_top) {
    }
}
```

項目	自動ICLNetworkVerify
TAP進入IJTAG network	會驗證
SIB可開啟到指定TDR	會為了存取TDR而操作
每個TDR可iWrite/iRead	手冊明確說明會驗證
所有 2^N 種SIB組合	不會窮舉
30-bit TDR每bit walking-1/0	不保證
valid一定在指定時間變1	不會理解此功能語意
ALU所有opcode與flags	不會; 需要application PDL
所有SDF corners/TCK頻率	不會; 必須分開執行

若valid是X但ICLNetworkVerify仍PASS, 先確認valid bit是否被mask.這可能代表scan path已通, 但時序instrument的valid功能尚未被測試.

7. 建議的四層驗證與最終清單

層級	目的	最低通過條件
L1 ICL transport	TAP/SIB/TDR access	ICLNetworkVerify與testbench status PASS
L2 TDR bits	bit order、scan length、stuck-at	0/1、AA/55、walking-1/0皆正確
L3 Instrument semantics	value/valid、event、clear、ALU operation	application PDL的expected read全部PASS
L4 Gate timing	製程cell與delay下可操作	gate no-SDF與指定SDF corners皆PASS, 無關鍵timing violation

valid X的實作檢查清單

- Gate/no-SDF是否仍會X?
- 第一個X是在event、shadow Q、TDR capture bit還是TDO mux?
- shadow register是否有正確reset, reset解除是否避開clock edge?
- event/clear是否跨functional clock與TCK domain?
- event與clear是否可能同時成立?
- Capture-DR前是否等待足夠functional clock與同步clock cycles?
- simulation log是否有shadow DFF的setup/hold/recovery/removal violation?
- valid對應的expected TDO bit是否被mask?
- netlist是否仍有shadow register, 且沒有被constant/merge optimization移除?

建議保存的證據

```
dc_check_design.rpt
dc_reference_and_register.rpt
gate_no_sdf_sim.log
gate_sdf_sim.log
sdf_annotation_summary.log
patterns_specification_report.txt
check_testbench_simulations_report.txt
waveform_first_x_timestamp.png
```

參考: Tessent Shell User's Manual v2021.3, Verifying the ICL Model (手冊頁157), Verifying ICL-Based Patterns After Synthesis (370-371), 以及 Synthesis Guidelines for RTL Designs with Tessent Inserted DFT (779-783).